

L11: Lesson Summary



This lesson focused on additional network dynamic processes – dynamics of networks (*failures and attacks on nodes*), on networks (*search and synchronization*), and coevolutionary network phenomena.

There are many more network dynamical processes that have been studied in the literature. We briefly mention some of them here:

- **Network control:** how to select the minimum set of nodes that, if externally controlled, can "drive" the network to the desired state?
- **Network densification:** are networks getting denser over time? If so, why and how?
- **Network congestion:** in networks where there is some flow of "traffic" and a limited capacity on each link (or node), how does congestion cascade from one bottleneck to the rest of the network?
- **Network games:** if we think of each node as a "rational agent", what happens to the entire network if each node optimizes its connections to maximize a "selfish" objective? (e.g., minimize the number of connections while being within a short distance from every other node)

In all these problems, the common question that network science focuses on is the following: how does the structure of the network affect the dynamic process that is taking place on the network? And if the network is adaptive, how does this dynamic process affect the structure of the network? The structure and the function of a network are always intertwined -- and the link between the two always relates to dynamic network processes.